

CS 490: Guided Design in Software Engineering

Martin Kellogg

Welcome to CS 490!

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Today's agenda:

- What is 490 + course policies and expectations
- About the instructor (aka why you should listen to me)
- In-class activity: background survey
- Survey of the project + other assignments (syllabus day!)
- Start “Code-level design” lecture (if time permits)

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Course policies

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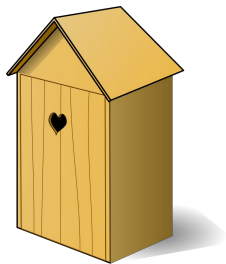
- **Most important:** the first time each class you ask or answer a question, I throw candy at you (sorry for poor aim)

Course policies

- **Most important:** the first time each class you ask or answer a question, I throw candy at you (sorry for poor aim)
 - Let's try it now! **Suggested questions:**
 - Why would you do that?
 - Are you just bribing us to pay attention?
 - Does that actually work?
 - Do even silly questions count?

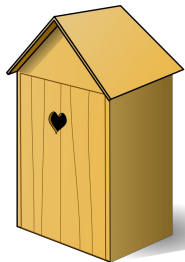
What is CS 490?

What is CS 490? An analogy



= CS 113/114

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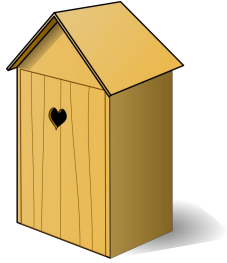


= CS 113/114



= CS 280/288

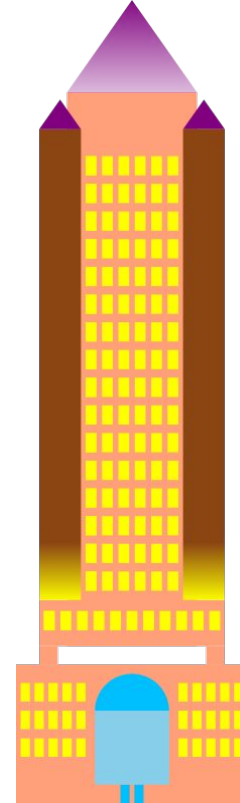
What is CS 490? An analogy



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= CS 490

What is CS 490?

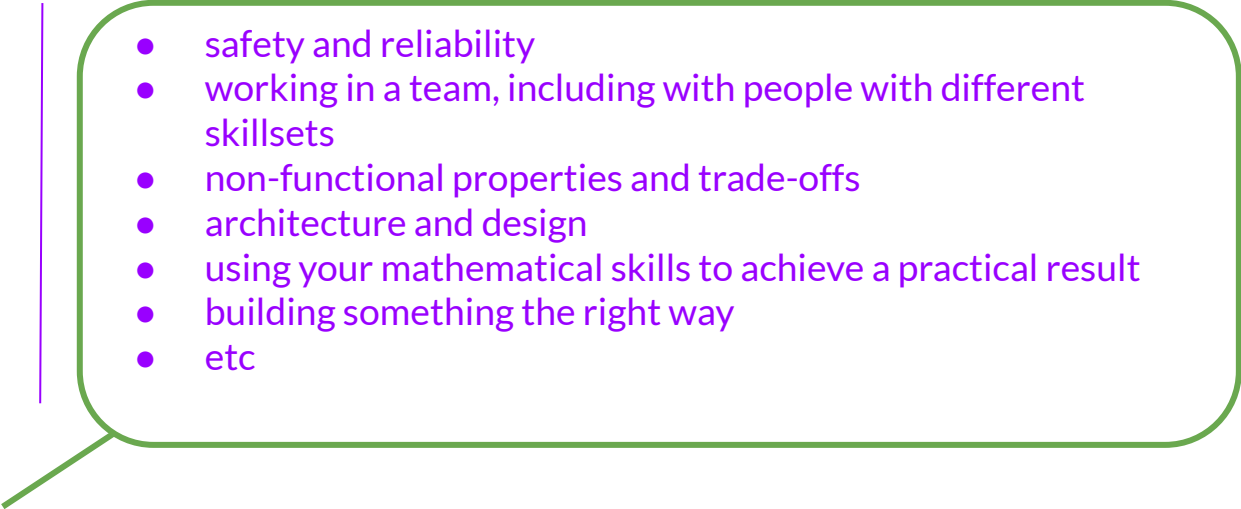
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- A course about *engineering* software

What is CS 490?

- Previous courses were about *programming*
- A course about *engineering* software
 - safety and reliability
 - working in a team, including with people with different skillsets
 - non-functional properties and trade-offs
 - architecture and design
 - using your mathematical skills to achieve a practical result
 - building something the right way
 - etc

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- safety and reliability
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 - etc

How do these principles apply to programming?

Why does software need to be engineered?

Why does software need to be engineered?

2023 FAA system outage

From Wikipedia, the free encyclopedia

(Redirected from [2023 FAA system outage in the United States](#))

On January 11, 2023, US flights were grounded or delayed as the [Federal Aviation Administration](#) (FAA) attempted to fix a system outage.^{[1][2]} FAA paused all flight departures until 9 a.m. ET.^[2] Flights already in the air were allowed to continue to their destinations.^[1] Around 8:30 a.m. ET, flights were beginning to resume departures.^[1] The outage was the first time since [September 11, 2001](#) that the FAA issued a nationwide [ground stop](#) in the United States.^[3]

A preliminary investigation of the incident demonstrated to FAA investigators that a "damaged database [file](#)" may have caused the outage of the FAA's Notice to Air Missions ([NOTAM](#)) system, responsible for notifying pilots of safety hazards.^[4] The FAA told [CNN](#) that there was "no evidence of a [cyberattack](#)" on its NOTAM system.^[4]

Why does software need to be engineered?

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Toyota Case: Single Bit Flip That Killed

Junko Yoshida

10/25/2013 03:35 PM EDT

During the trial, embedded systems experts who reviewed Toyota's electronic throttle source code testified that they found Toyota's source code defective, and that it contains bugs -- including bugs that can cause unintended acceleration.

"We did a few things that NASA apparently did not have time to do," Barr said. For one thing, by looking within the real-time operating system, the experts identified "unprotected critical variables." They obtained and reviewed the source code for the "sub-CPU," and they "uncovered gaps and defects in the throttle fail safes."

The experts demonstrated that "the defects we found were linked to unintended acceleration through vehicle testing," Barr said. "We also obtained and reviewed the source code for the black box and found that it can record false information about the driver's actions in the final seconds before a crash."

Stack overflow and software bugs led to memory corruption, he said. And it turns out that the crux of the issue was these memory corruptions, which acted "like ricocheting bullets."

Barr also said more than half the dozens of tasks' deaths studied by the experts in their experiments "were not detected by any fail safe."

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http://www.eetimes.com/document.asp?doc_id=1319903&page_number=1
(excerpts)

"Task X death
in combination
with other task
deaths"

Why does software need to be engineered?

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The experts determined that the unintended acceleration also occurred. They found that it could occur in the final seconds of the race.

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Barr also said that the experts in the field were "safe."

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Reference ID: 0.cdd74f17.1380634949.2f9c301c

Health Insurance Marketplace

181 DAYS LEFT TO ENROLL

OCT 1 Open Enrollment Begins

JAN 1 Coverage Can Begin

MAR 31 Open Enrollment Closes

Live Chat

Why does software need to be engineered?

2023 FAA system outage

From Wikipedia, the free encyclopedia

Ariane flight V88^[1] was the failed maiden flight of the **Arianespace Ariane 5** rocket, vehicle no. 501, on 4 June 1996. It carried the **Cluster** spacecraft, a constellation of four **European Space Agency** research satellites.

The launch ended in failure due to multiple errors in the software design: **dead code**, intended only for **Ariane 4**, with inadequate protection against **integer overflow** led to an **exception handled** inappropriately, halting the whole otherwise unaffected **inertial navigation system**. This caused the rocket to veer off its flight path 37 seconds after launch, beginning to disintegrate under high aerodynamic forces, and finally self-destructing via its automated **flight termination system**. The failure has become known as one of the most infamous and expensive **software bugs** in history.^[2] The failure resulted in a loss of more than US\$370 million.^[3]

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The **Therac-25** was a computer-controlled radiation therapy machine produced by Atomic Energy of Canada Limited (AECL) in 1982 after the Therac-6 and Therac-20 units (the earlier units had been produced in partnership with *Compagnie Générale de Radiologie* (CGR) of France).

It was involved in at least six accidents between 1985 and 1987, in which patients were given massive overdoses of radiation.^{[1]:425} Because of concurrent programming errors (also known as race conditions), it sometimes gave its patients radiation doses that were hundreds of times greater than normal, resulting in death or serious injury.^[2] These accidents highlighted the dangers of software control of safety-critical systems, and they have become a standard case study in health informatics, software engineering, and computer ethics. Additionally, the overconfidence of the engineers^{[1]:428} and lack of proper due diligence to resolve reported software bugs are highlighted as an extreme case where the engineers' overconfidence in their initial work and failure to believe the end users' claims caused drastic repercussions.

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Live Chat

Why does software need to be engineered?

The code you write will
have ✨*consequences* ✨
in the real world!

2023 FAA

From Wikipedia,

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My expectations

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 - you can write code
 - you can program against an English specification
 - you can read code and figure out what it does
 - you can teach yourself a new programming language
 - you can debug code that's not behaving like you expect
 - you can install software yourself + do basic troubleshooting
 - when you get stuck, you know how to google/chatgpt for answers

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The ability to solve problems yourself
with a search engine or LLM is a **critical**
skill for a software engineer!

My expectations

- You know how to *program*
- Professionalism

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- You know how to *program*
- Professionalism
- Participation

CS 490 goals

Officially the following:

- Students will be able to explain the major theories and methods applicable to professional software engineering.
- Students will be able to design, implement and evaluate a computer based system to meet desired needs.
- Students will be able to function effectively on a team to accomplish a goal.
- Students will be able to use current techniques, skills and tools necessary for computing practice.

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course project!

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My goals for you:

- Students will be able to assess the **quality of software engineering** being done at some future workplace
- Students will be **competent software engineers** that I wouldn't be worried about hiring

CS 490 goals: What about AI?

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 - I think this talk is **mostly wrong**, for two reasons
- Reason 1: software engineering is about **more than writing code**
 - e.g., researchers at Microsoft in 2017 found that devs spent only about 17% of their time in their IDE¹

¹ Meyer, A. N., Barton, L. E., Murphy, G. C., Zimmermann, T., & Fritz, T. (2017). The Work Life of Developers: Activities, Switches and Perceived Productivity. IEEE Transactions on Software Engineering, 43, 1178–1193. <https://doi.org/10.1109/tse.2017.2656886>

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- Reason 1: software engineering is about **more than writing code**
 - e.g., researchers at Microsoft in 2017 found that devs spent only about 17% of their time in their IDE¹
- Reason 2: the **Jevons paradox**² (an idea from economics)
 - As a good becomes cheaper, demand for that good increases more than proportionally, leading to higher total use

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² https://en.wikipedia.org/wiki/Jevons_paradox

CS 490 goals: What about AI?

- There's been a lot of talk in the press lately about "AI replacing software engineers"
 - Jevons Paradox applied to AI: software is getting cheaper, so it'll be cost effective to use software for more things. So, it's likely total demand for software will rise. So, we'll need **more** software engineers!
- Real world example: AI writing code
 - At devs spent
- Real world example: AI writing code (economics)
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I'm an academic, not a professional software engineer



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- My research area is in software engineering
 - more specifically, static analysis design (“compilers”)
- ~25% of my PhD spent embedded at AWS
 - two co-authored publications
 - my analysis tools deployed on > 70M lines of AWS code
- My lab is one of the few in the world to take SE seriously when writing research code
 - Inherited from my PhD advisor, who employed 3 SDEs concurrently while I was a student!

Our TAs

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 - they know covey.town well and have experienced this course “in your shoes”

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- TA office hours are where you should go for help with covey.town
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 - if you have questions about lecture contents, my OH are best
- You can also ask on Discord
 - both the TAs and I will be monitoring Discord, so you can ask any questions you'd like there

Coordination

- 490 is (sort of) **coordinated** this semester:
 - this section and section 003 (Mohit Dale's) will share assignments, TAs, exams, office hours, Discord, etc
 - you can go to either Mohit or I with any concerns you have

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 - you can go to either Mohit or I with any concerns you have
- Jeri Qerimaj is also shadowing me this semester
 - the plan is that she will start teaching from this curriculum, too, going forward
 - she'll also have office hours that any of you can use
 - Tuesday and Thursday 10:00 am - 11:20 am, GITC 4301

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Break: background survey

<https://forms.gle/nuAu2T9HrR96Bumz6>



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A brief tour through the course website

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This website is the *single source of truth* in this class. If, e.g., Canvas says a due date is “Time X” but the website says “Time Y”, **assume the website is correct.**

A brief tour through the course website

- <https://web.njit.edu/~mjk76/teaching/cs490-au26/>
 - Mandatory readings + reading quizzes
 - “Your Choice” readings
 - Individual Project 0: due < 1 week from today
 - My grading: “tough but fair” + curve at the end
 - Collaboration policy (I expect you to use a search engine!)
 - Project structure
 - How to get help
 - Overview of topics

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Action items before next week

- Individual Project 0
- Mandatory readings for next week's classes: there will be quizzes!
 - Syllabus and IPO specifications ("today's readings") are also fair game
- Note that this class **meets on Tuesday and Wednesday next week**
- Make sure you can access all course materials
 - Course website
 - Canvas
 - Discord